7
Real and Ideal P.I.D Controllers

7.1 Objectives

As a result of studying this chapter, and after having completed the relevant exercises, the student should be able to:

- Select the correct PID-Control algorithm for field interaction and for computer optimized calculations;
- Clearly distinguish between process noise and control loop instability, which are often similar in appearance;
- List the correct sequence of steps to handle the different problems of noise and instability.

7.2 Comparative Descriptions of Real and Ideal Controllers

The **Ideal PID-Controller** is not suitable for direct field interaction, therefore it is called the **Non-Interactive** PID-Controller. It is highly responsive to electrical noise on the PV input if the Derivative function is enabled.

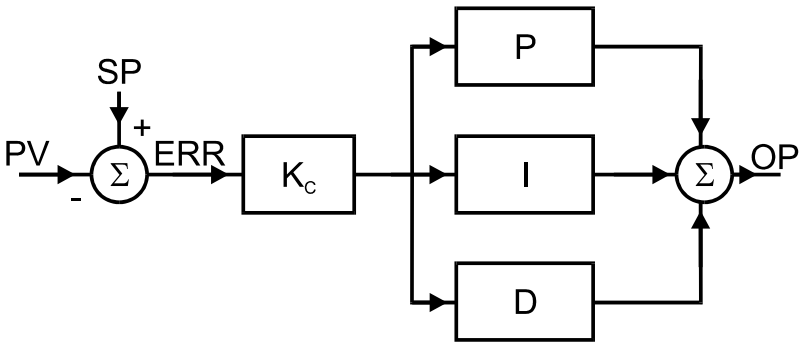
The **Real PID-Controller** is especially designed for direct field interaction, is therefore called the **Interactive** PID-Controller. Due to internal filtering in the Derivative block the effects of electrical noise on the PV input is greatly reduced.

7.3 Description of the IDEAL or the Non-Interactive PID Controller

The Non-Interactive form of controller is the classical teaching model of PID algorithms. It gives a student a clear understanding of P, I and D control, since:

P-Control I-Control and D-Control can be seen independently of each other.

Then, PID is effectively a combination of independent P, I and D-control actions. This can be seen in Figure 7.1.



$$OP = K_C \times \left(T_{der} + \frac{1}{T_{int}} + 1 \right)$$

OP = Gain × (D-Control + I-Control + P-Control)

Figure 7.1
Ideal PID-Controller

Since P, I and D algorithms are calculated independently in an Ideal PID-Controller, this form of controller is recommended if an ***ideal process variable exists.***

7.3.1 Ideal Process Variables

An ideal process variable is a noise-free, refined and optimized variable. They are a result of computer optimization, process modeling, statistical filtering and value prediction algorithms.

These types of ideal process variables do not come from field sensors.

In these cases, it is of great benefit that the actual formula of the ***Ideal PID algorithm*** is simple, as shown in Figure 7.2 below:

$$OP = Kc * \left(T_{der}\, s + \frac{1}{T_{int}\, s} + 1 \right)$$

OP = Gain * (D-Control + I-Control + P-Control)

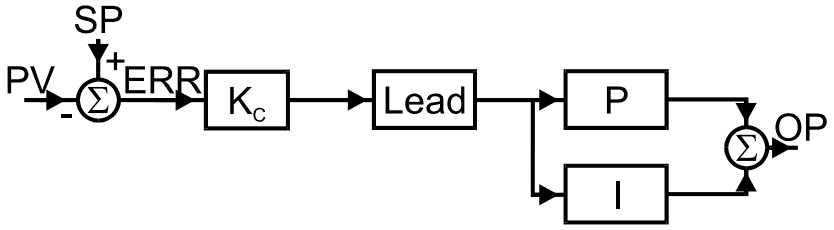
Figure 7.2
Ideal PID Algorithm

7.4 Description of the Real (Interactive) PID Controller

The Interactive form is the PID algorithm used for direct field control. That is either or both of its input (PV) and output (MV) are directly connected to field or process equipment. It is designed to cope with any electrical noise induced into its circuits by equipment in the plant or factory.

Full understanding of the Interactive PID algorithm is rather difficult, since P-Control, I-Control and D-Control cannot be seen independently from each other. Therefore, Interactive PID is not just a sum of independent P, I and D control.

This can be seen in Figure 7.3 below:



$$OP = K \times \frac{T_{der} + 1}{\alpha T_{der} + 1} \times \frac{1 + T_{int}}{T_{int}}$$

$$OP = Gain \times Lead \times PI\text{-}Control$$

Figure 7.3
Real PID-Controller

Since the Interactive PID-Controller makes use of a Lead algorithm rather than using the classical mathematical derivative, it is best suited for real (field) process variables.

7.4.1 Real Process Variables (Field Originated)

A real process variable has electrical noise that come from field sensors or the connecting cables.

It is therefore of great benefit that the PID algorithm has some noise reduction built in.

The formula below represents an Interactive PID algorithm:

$$OP = K_c * \frac{T_{der}s + 1}{\alpha T_{der}s + 1} * \frac{1 + T_{int}s}{T_{int}s}$$

OP = Gain * Lead * PI-Control

Figure 7.4
Real PID Algorithm

7.5 Lead Function - Derivative Control with filter

The following is an extract from chapter 6 (Digital Control Principles) to remind us of the LEAD part acting as a **Derivative function.**

The field controller uses a Lead Algorithm for derivative control. The block diagram is shown in Figure 7.5.

7.5.1 Lead Algorithm for Derivative Control (Field or Real PID Controller)

$$\frac{Ts + 1}{\alpha Ts + 1} = \left(\frac{1}{\alpha Ts + 1}\right) \text{ x } (Ts + 1)$$

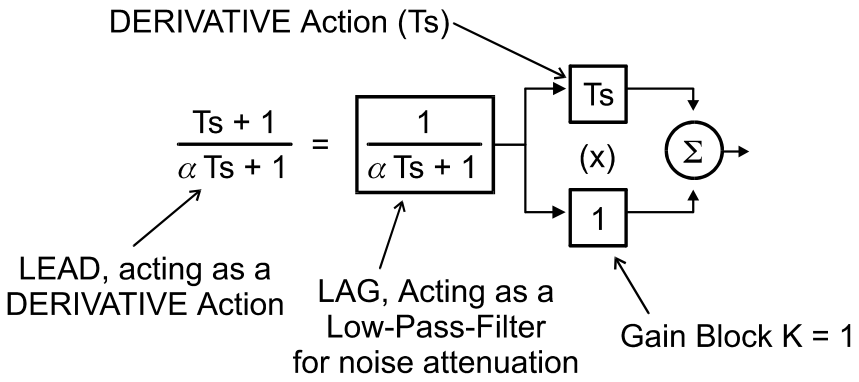


Figure 7.5
Block Diagram of Lead as Derivative

7.6 Derivative Action and Effects of Noise

The most important difference between **NON-INTERACTIVE AND INTERACTIVE PID CONTROLLERS** is the different impact noise has on a controller's output.

It must be remembered that derivative control multiplies noise.

7.6.1 Introduction to Filter Requirements

Both Non-Interactive PID and Interactive PID controllers make use of a noise filter for process noise (***known as the process variable filter time constant TD***).

Since the derivative control of a Non-Interactive PID has no noise suppression of its own, noise will always be a major problem, even though a Process Variable filter may be used.

Since the derivative control of an Interactive PID already has some noise suppression of its own, noise is not so much of a problem, and is even less if a process variable filter is used.

It is recommended that a PV filter should be used in all cases where derivative control is being used. The author has observed numerous derivative control systems having excessive movement of the controller outputs due to the lack of PV filters.

This type of problem is often incorrectly interpreted by personnel (in industrial plants) as being a problem of stability.

Hence an important rule is:

Make a clear distinction between noise and instability in industrial control applications.

As discussed earlier, noise and instability require treatment with different methodologies, as they are totally different problems.

Remember, a PROCESS VARIABLE FILTER, due to its LAG action, reduces noise but may add to loop instability.

EXERCISE 5 INTRODUCTION TO DERIVATIVE CONTROL
On the subject of D-Control in Non-Interactive and Interactive PID-Controllers and the significance of noise.

7.7 Example of the KENT K90 Controllers PID Algorithms

Proportional Control = $K1 = \frac{100}{PB}$

Integral Control = $K2 = \frac{100}{PB} \times \frac{\text{Scan Period}}{\text{IAT(Sec.)}}$

Derivative Control = $K3 = \frac{100}{PB} \times \frac{\text{DAT(Sec.)}}{\text{Scan Period}}$

$$\text{Proportional} = K1 \times \text{Error}$$

$$\text{Integral} = I_{N-1} + \frac{K2(SP_N - MV_N)}{\text{MV Range}}$$

$$\text{Derivative} = \frac{K3(MV_{N-1} - MV_N)}{\text{MV Range}}$$

Result = Proportional + Integral + Derivative

$$= (K1 \times \text{Error}) + \left(I_{N-1} + \frac{K2\,(SP_N - MV_N)}{\text{MV Range}}\right) + \left(\frac{K3(MV_{N-1} - MV_N)}{\text{MV Range}}\right)$$

$$= \frac{100}{PB}\left(E + \frac{1}{IAT}\int E.dt + DAT.\frac{dPV}{dt}\right)$$